	Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Robotics & Artificial Intelligence Department		
Subject Name	**Computer Organization and Microprocessor (COMA)**	**A.Y**	**2025-26**
Subject Code	**303105210**	**Semester**	**4th**

Sr No	Question	COs	B.T	Competence
Q.1	Answer the following Questions (1 Marks questions)			
1.1	How many maximum memory locations and I/O devices can be addressed by an 8085 Microprocessor?	CO1	BT-1	Remember
1.2	Explain the use of HOLD and HLDA pins of 8085 microprocessor.	CO1	BT-2	Understand
1.3	State the difference between op-code fetch (OF) and memory read (MR) cycles.	CO1	BT-4	Analyse
1.4	State the importance of X1 and X2 pins of an 8085 microprocessor.	CO1	BT-1	Remember
1.5	What is an interrupt? Enlist the hardware interrupt sources (pins) available on the 8085 Microprocessor chip.	CO3	BT-1	Remember
1.6	Explain the use of READY pin of an 8085 microprocessor.	CO3	BT-2	Understand
1.7	Explain RESET-IN AND RESET-OUT pin of an 8085 microprocessor.	CO1	BT-2	Understand
1.8	What is Instruction Cycle?	CO1	BT-1	Remember
1.9	What is Machine Cycle?	CO1	BT-1	Remember
1.10	Which interrupt has the highest priority of 8085 Microprocessor?	CO3	BT-1	Remember
1.11	What is SIM?	CO3	BT-1	Remember
1.12	Calculate address line for 4, 8, 16 byte memory.	CO3	BT-3	Apply
1.13	What does microprocessor speed depends	CO1	BT-1	Remember

	on?			
Q.2	Describe the following questions. (2-3 Marks questions)			
2.1	Explain Flag Register of 8085 Microprocessor.	CO1	BT-2	Understand
2.2	Explain Controlling signal generating logic circuit.	CO1	BT-2	Understand
2.3	Describe the status signal (S0 and S1) for Memory and I/O operation.	CO1	BT-2	Understand
2.4	Explain the function of the ALE and IO/M signal of the 8085 microprocessor.	CO1	BT-2	Understand
2.5	Explain the need to De-multiplex the bus AD7-AD0.	CO1	BT-2	Understand
2.6	If the clock frequency if 5Mhz, how much time is required to execute an instruction of 18 T-States?	CO2	BT-3	Apply
2.7	Explain the structure of RAM and ROM.	CO1	BT-2	Understand
2.8	Explain Memory Read operation.	CO1	BT-2	Understand
2.9	Explain Memory Write operation.	CO1	BT-2	Understand
2.10	Which is the instruction used for increment and decrement purpose? Give example of it.	CO2	BT-1	Remember
Q.3	Describe the following questions. (7-8 Marks).			
3.1	Draw the functional block diagram of internal architecture of IC 8085 and explain its working.	CO1	BT-2	Understand
3.2	Draw the Pin diagram of 8085 Microprocessor and explain each pin.	CO1	BT-2	Understand
3.3	Explain Memory interfacing with necessary steps and diagram.	CO3	BT-6	Create
3.4	What is the function of ALE pin in 8085?	CO1	BT-2	Understand
3.5	Draw the memory interfacing of the 8K byte EPROM and 8K byte RAM with starting address 6000H and 7000h respectively.	CO3	BT-6	Create
3.6	Draw the memory interfacing of the 2K byte EPROM and 1K byte RAM.	CO3	BT-6	Create
3.7	Explain the logical instruction in detail.	CO2	BT-2	Understand
3.8	Can you describe how the addition instruction with carry operates?	CO2	BT-2	Understand

3.9	Which are the control instructions in 8085? Discuss it.	CO2	BT-4	Analyse